Applied Math and Machine Learning Basics (2)

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Special Lectures for Machine Learning

Outline

Probability I

1. Mathematical def.
   - A set function $\mathbb{P} : \sigma(\Omega) \to [0, 1]$ where Ω is a sample space.
   - Axioms
     1. $\mathbb{P}(\Omega) = 1$.
     2. $\mathbb{P}(A) \geq 0$ where $A \in \sigma(\Omega)$.
     3. $\mathbb{P}(\sum_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$ for all A_i such that $A_i \in \sigma(\Omega),\ A_i \cap A_j = \phi (i \neq j)$.
   - Probability is used to measure the uncertainty. What's the source of uncertainty?

Probability II

2. Source of uncertainty and interpretation
   - Source of uncertainty
     1. Inherent stochasticity in the system being modeled. For example, quantum mechanics.
     2. Incomplete observability. Existence of latent variable.
     3. Incomplete modeling. Discarding observed information in the modeling.
   - Interpretation of "probability"
     1. Frequentist: the rates at which events occur in the infinite trials.
     2. Bayesian: degree of belief
     - In the real world, repeated events or non-repeated events exist simultaneously.

Probability III

3. Probability distributions

- Random variable X: mapping $\Omega \to \mathcal{R}$.

$$\{X \in A\} \equiv \{w \in \Omega : w \in X^{-1}(A)\}.$$

- Distribution function $F(x) = \mathbb{P}(X \leq x),\ x \in \mathbb{R}$.
- Support: The closure of a minimal set having the probability of one.
- Discrete random variables and probability mass function
  1. $p(x) = \mathbb{P}(X = x) = F(x) - F(x-)$.
  2. $\sum_{x \in \mathcal{X}} p(x) = 1$.
  3. $p(x, y) = \mathbb{P}(x = x, Y = y)$.
- Examples of discrete random variables:
  1. Uniform : $p(x) = \frac{1}{|\mathcal{X}|}$.
  2. Poisson : $p(x) = \frac{\lambda^x \exp(-\lambda)}{x!}, x \in \{0\} \cup \mathbb{N}$ where $\lambda > 0$.

Probability IV

- Continuous random variables probability density functions
  1. $\mathbb{P}(X \in [a,b]) = \int_{[a,b]} p(x)dx$.
  2. Here, 1) $p(x) \geq 0$, and 2) $\int_{\mathcal{X}} p(x) = 1$.
- Notes:
  1. In fact, the conti. means the support of a random variable is infinitely uncountable.
  2. Density function cannot be unique, and the probability mass function can be considered as the density function because

$$\sum_{x \in \mathcal{I}} p(x) = \sum_{x \in \mathcal{X} \cap \mathcal{I}} p(x) = \int_I p(x)\mu(dx)$$

     where $\mu(dx)$ is a counting measure.

Joint Probability I

1. Marginal and conditional probability
   - Marginal prob.
     1. Discrete case

$$\forall x \in \mathcal{X}, p(x) = \sum_{y \in \mathcal{Y}} p(x, y)$$

        where the support of (X, Y) is $\mathcal{X} \times \mathcal{Y}$.

     2. Continuous case

$$\forall x \in \mathcal{X}, p(x) = \int_{\mathcal{Y}} p(x, y) dy.$$

Joint Probability II

- Conditional probability
  1. $\mathbb{P}(X = x \mid Y = y) = \frac{\mathbb{P}(X=x, Y=y)}{\mathbb{P}(Y=y)}$.
  - Different to the action in causal inference.
  2. When $\mathbb{P}(Y = y) = 0$, then conditional probability is defined as $g(x \mid y) \equiv g(x, y)$ satisfying:

$$\mathbb{P}(X \in \mathcal{V}, Y \in \mathcal{W}) = \int_{\mathcal{W}} \left[\int_{\mathcal{V}} g(x, y) p(y) dy \right] dx$$

for any measurable $\mathcal{V} \in \mathcal{Y}$ and $\mathcal{W} \in \mathcal{X}$.

Joint Probability III

2. Chain rule for conditional probabilities and independence
   - Chain rule

$$\begin{aligned}\mathbb{P}(x^{(1)}, x^{(2)}, \ldots, x^{(n)}) &= \mathbb{P}(x^{(1)}) \prod_{i=2}^{n} \mathbb{P}(x^{(i)} \mid x^{(i-1)}, \ldots, x^{(1)}) \\ &= \mathbb{P}(x^{(n)}) \prod_{i=1}^{n-1} \mathbb{P}(x^{(i)} \mid x^{(i+1)}, \ldots, x^{(n)})\end{aligned}$$

   - Independence with two r.v.s, $X \perp Y$

$$\forall x, y,\ \mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y)$$

   - Mutual indep.: Assume $X_i, i = 1, \ldots, n$,

$$\mathbb{P}(\cap_{i \in \mathcal{P}} X_i \in A_i) = \prod_{i \in \mathcal{P}} \mathbb{P}(X_i \in A_i),$$

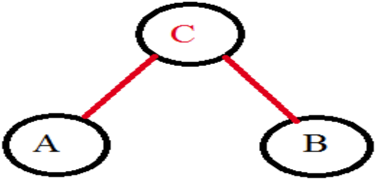
where $\mathcal{P}$ is any subset of $\{1, \ldots, n\} = [n]$.

Joint Probability IV

- Conditional indep.: $X \perp\!\!\!\perp Y \mid Z$

$$\mathbb{P}(X = x, Y = y \mid Z = z) = \mathbb{P}(X = x \mid Z = z)\mathbb{P}(Y = y \mid Z = z)$$

1. For example, Z is the subject number and X and Y are math and physics sores.
2. Then, math and physics scores can be correlated due to the other subjects. However, given the subject, math and physics scores are not correlated.



Joint Probability V

3. Expectation, variance, and covariance

- Expectation

$$\mathbb{E}_{X\sim p}[x] = \int_{\mathcal{X}} xp(x)\mu(dx).$$

1. $\mathbb{E}_{(X,Y)\sim p}[\alpha X + \beta Y] = \alpha \mathbb{E}_{X\sim p}[X] + \beta \mathbb{E}_{Y\sim p}[Y]$
2. $\mathbb{E}_{(X,Y)\sim p}[XY] = \mathbb{E}_{X\sim p}[X]\mathbb{E}_{Y\sim p}[Y]$ does not hold in general. Equality holds when $X \perp Y$.

Joint Probability VI

- Variance and covariance
  1. $\mathrm{Var}(f(x)) = \mathbb{E}\left[(x - \mathbb{E}[f(x)])^2\right].$
  2. $\mathrm{Cov}(f(x), g(y)) = \mathbb{E}\left[(f(x) - \mathbb{E}[f(x)])(g(y) - \mathbb{E}[g(y)])\right].$

Figure 1: https://www.scribbr.com/statistics/standard-deviation/

Joint Probability VII

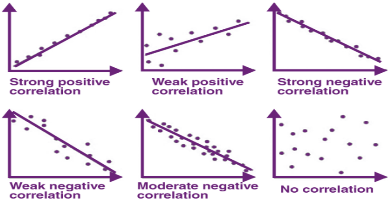


Figure 2: https://byjus.com/maths/correlation/

- The independence implies the zero covariance, but its reverse is not true.

Common Distributions I

1. Bernoulli
   - $\mathbb{P}(X = x \mid \phi) = \phi^x (1-\phi)^{1-x}, 0 < \phi < 1$ where $\mathcal{X} = \{0, 1\}$.
   - $\mathbb{E}[X] = \phi$.
   - $\mathrm{Var}(X) = \phi(1-\phi)$.
2. Multinoulli
   - $\mathbb{P}(X = x \mid \boldsymbol{p}) = \prod_{k=1}^{K} {p_k}^{\mathbb{I}(x_k=1)}, 0 < p_k < 1$ $(\sum_k p_k = 1)$ where $\mathcal{X} = \{0, 1\}^K$ with a restriction $\sum_k x_k = 1$ for $x \in \mathcal{X}$.

Common Distributions II

3. Gaussian
   - $p(x \mid \mu, \sigma) = (2\pi\sigma^2)^{-1/2} \exp(-0.5(x-\mu)^2/\sigma^2)$ where $\mu \in \mathbb{R}, \sigma > 0$.
   - $\mathbb{E}[X] = \mu, \mathrm{Var}(X) = \sigma^2$
   - Let $\beta = 1/\sigma^2$, referred to as a precision.
   - Merits: Central limit theorem, the maximum amount of uncertainty among distributions with the same variance.

Common Distributions III

4. Multivariate normal distribution
   - $p(\boldsymbol{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \det(2\pi\boldsymbol{\Sigma})^{-1/2} \exp(-0.5(\boldsymbol{x} - \boldsymbol{u})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x} - \boldsymbol{u}))$ where $\boldsymbol{\mu} \in \mathbb{R}^d, \boldsymbol{\Sigma} \succ 0$.
   - $\boldsymbol{\beta} = \boldsymbol{\Sigma}^{-1}$ is a precision matrix.
   - Isotropic Gaussian: covariance matrix is a scalar times the identity matrix.
4. Exponential and Laplace
   - $p(x \mid \lambda) = \lambda \exp(-\lambda x)\mathbb{I}(x \geq 0)$.
   - $p(x \mid \mu, \gamma) = (0.5/\gamma) \exp(-|x - \mu|/\gamma)$.

Common Distributions IV

5. Dirac and empirical
   - $p(x \mid \ \mu) = \mathbb{I}(x = \mu)$, not general (limit distribution having the peak at μ.)
   - $p(x \mid \{x^{(i)}\}_{i=1}^{m}) = \sum_{i=1}^{m} \frac{1}{m}\mathbb{I}(x = x^{(i)})$ where $\{x^{(i)}\}_{i=1}^{m}$ is a random samples. Using the empirical frequency.
   - Sample distribution via training random samples and maximum likelihood estimator for probability density function.

Common Distributions V

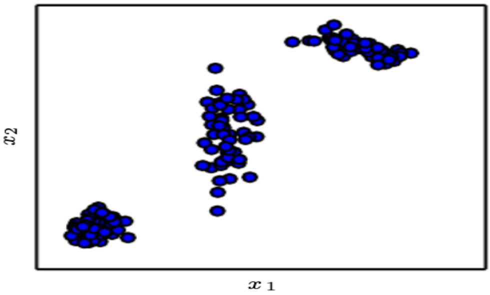
6. Mixture of distributions

- $p(x) = \sum_i p(c = i)p(x \mid c = i)$.
- Usually, c is considered as a (unobserved) latent variable.
- Gaussian mixture

$$p(\boldsymbol{x} \mid \boldsymbol{\alpha}, \boldsymbol{\mu}^{(1)}, \ldots, \boldsymbol{\mu}^{(n)}, \boldsymbol{\Sigma}^{(1)}, \ldots, \boldsymbol{\Sigma}^{(n)},) = \sum_i \alpha_i p(\boldsymbol{x} \mid \boldsymbol{\mu}^{(i)}, \boldsymbol{\Sigma}^{(i)})$$

- Prior probability α_i and posterior probability $p(c = i \mid \boldsymbol{x})$.
- Gaussian mixture is a universal approximator for densities.

Common Distributions VI



Common Distributions VII

7. Some properties
   - Logistic sigmoid: $\sigma(x) = 1/(1 + \exp(-x))$, having nearly no change when the $\boldsymbol{x}$ is larger or very smaller (saturated).
   - Softplus function: $\zeta(x) = \log(1 + \exp(x))$, soft-end version of $x^+ = \max(x, 0)$.

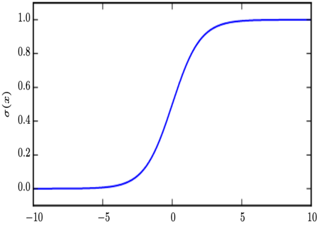


Figure 3.3: The logistic sigmoid function.

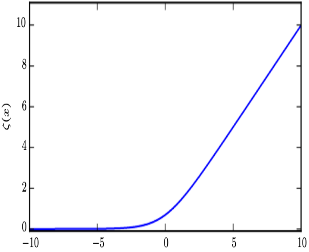


Figure 3.4: The softplus function.

Common Distributions VIII

- Properties
  1. $\frac{d\sigma(x)}{dx} = \sigma(x)(1 - \sigma(x))$.
  2. $1 - \sigma(x) = \sigma(-x)$.
  3. $\log \sigma(x) = -\zeta(-x)$.
  4. $\frac{d\zeta(x)}{dx} = \sigma(x)$.
  5. $\zeta(x) - \zeta(-x) = x$.

Bayes' Rule I

Bayes' rule

- $p(y \mid x) = \frac{p(x|y)p(y)}{p(x)}$

- $p(y)$: prior
- $p(x \mid y)$: likelihood
- $p(x)$: marginal prob.

Figure 3: Thomas Bayes (1701–1761)

Measure Theory and Jacobian I

Lebesgue measure

- It is a measure for the conti. r.v.s
- It gives 0 mass to countable points in the real line.
- For $\mathbb{R}^d$, it gives positive values to a non-zero volumes in the d-dimensional real.
- "Almost everywhere" holds throughout all space except for on a set of measure 0.

Measure Theory and Jacobian II

Jacobian

- For a random variable X, $g(X)$ has the probability such that $\mathbb{P}(g(X) \in A) = \mathbb{P}(X \in g^{-1}(A))$ where g is continuous (differentiable) function.
- If the density of X is f, what's the density of $g(X)$?

$$\begin{aligned} |p_y(g(x))dy| &= |p_x(x)dx| \\ p_y(y) &= p_x(g^{-1}(y))\left|\frac{dx}{dy}\right| \end{aligned}$$

or

$$p_x(x) = p_y(g(x))\left|\frac{dg(x)}{dx}\right|.$$

- For random vectors, we have

$$p_{\mathbf{x}}(g^{-1}(\mathbf{y}))\left|\det\frac{\partial \mathbf{x}}{\partial \mathbf{y}}\right|.$$

Information Theory I

Background

- It was originally invented to study sending messages from discrete alphabets over a noisy channel, such as communication via radio transmission.
- This field is fundamental to many electrical engineering and computer science areas.

Quantify information

- Likely events should have low information content.
- Less likely events should have higher information content.
- Independent events should have additive information.

Information Theory II

Self-information and Shannon entropy

- Self-information of event $X = x$: $I(x) = \log P(x)$
- Shannon entropy

$$H(\mathbf{x}) = -\mathbb{E}_{\mathbf{x}\sim P}[\log P(x)]$$

1. Expected amount of information in an event drawn from that distribution.
2. When the $\mathbf{x}$ is continuous, it is called as *differential entropy.*

Information Theory III

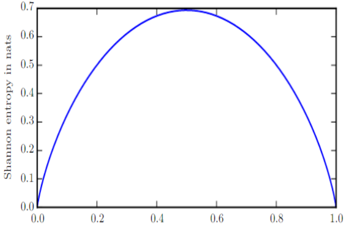


Figure 3.5: Shannon entropy of a binary random variable. This plot shows how distributions that are closer to deterministic have low Shannon entropy while distributions that are close to uniform have high Shannon entropy. On the horizontal axis, we plot p, the probability of a binary random variable being equal to 1. The entropy is given by $(p-1)\log(1-p) - p\log p$. When p is near 0, the distribution is nearly deterministic, because the random variable is nearly always 0. When p is near 1, the distribution is nearly deterministic, because the random variable is nearly always 1. When $p = 0.5$, the entropy is maximal, because the distribution is uniform over the two outcomes.

Figure 4: Shannon entropy of Bernoulli r.v. with p.

Information Theory IV

- Kullback-Leibler (KL) divergence of P and Q

$$\mathrm{KL}(P\|Q) = \mathbb{E}_{\mathbf{x}\sim P}\left[\log \frac{P(x)}{Q(x)}\right]$$

1. Non-negative and 0 if and only if $P = Q$ almost everywhere.
2. It is not true that $\forall P, Q, \mathrm{KL}(P\|Q) \neq \mathrm{KL}(Q\|P)$.

Information Theory V

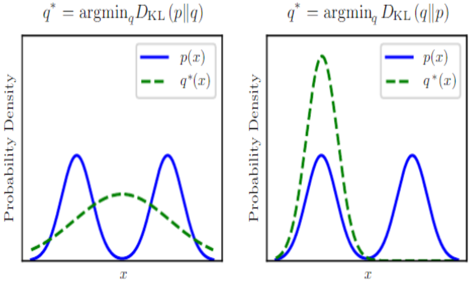


Figure 5: Example of KL divergence.

Information Theory VI

- Mutual information

$$\begin{aligned} I(X;Y) &= \mathrm{KL}(\mathbb{P}_{(X,Y)} \| \mathbb{P}_X \otimes \mathbb{P}_Y) \\ &= \int_{\mathcal{X}} \int_{\mathcal{Y}} p(x,y) \log \frac{p(x,y)}{p(x)p(y)} d\mu(y) d\mu(x). \end{aligned}$$

1. $X \perp Y \Leftrightarrow I(X;Y) = 0$.
2. $I(X;Y) = I(Y;X)$.
3. $I(X;Y) = H(Y) - H(Y \mid X) \geq 0$. It is interpreted as that the mutual information is the decrease by the conditional on X.

Information Theory VII

Structured probabilistic models

- Motivation: The computation burden is large when considering the probability of many objects.

$$\mathbb{P}(a_1, \ldots, a_k) = \prod_{i=1}^{k} \mathbb{P}(a_i \mid a_{i-1}, \ldots, a_1)$$

- If we can let $\mathbb{P}(a_i \mid a_{i-1}, \ldots, a_1) = \mathbb{P}(a_i \mid \mathcal{A}_i)$ where $\mathcal{A}_i \subset \{a_{i-1}, a_{i-2}, \ldots a_1\}$, we can greatly reduce the cost of representing a distribution.
- When we represent the factorization of a probability distribution with a graph, we call it a structured probabilistic or graphical model.

Information Theory VIII

- Two types: directed and undirected.
- **Directed:** having the parents represented by the directed graph.

$$p(\mathbf{x}) = \prod_i p(x_i \mid Pa_{\mathcal{G}\,i})$$

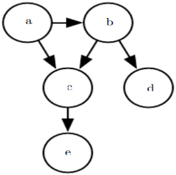


Figure 3.7: A directed graphical model over random variables a, b, c, d and e. This graph corresponds to probability distributions that can be factored as

$$p(\mathrm{a}, \mathrm{b}, \mathrm{c}, \mathrm{d}, \mathrm{e}) = p(\mathrm{a})p(\mathrm{b} \mid \mathrm{a})p(\mathrm{c} \mid \mathrm{a}, \mathrm{b})p(\mathrm{d} \mid \mathrm{b})p(\mathrm{e} \mid \mathrm{c}). \qquad (3.54)$$

Figure 6: Example of a directed graphical model.

Information Theory IX

- **Undirected:** Unusual to directed.
  - Any set of nodes that are all connected to each other in $\mathcal{G}$ is called a clique. Each clique $C(i)$ in an undirected model is associated with a factor $\phi^{(i)}(C(i))$.
  - $\phi^{(i)}(C(i))$ is not a probability

$$p(\mathbf{x}) = \frac{1}{Z}\prod_i \phi^{(i)}(C(i))$$

Information Theory X

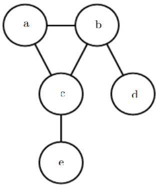


Figure 3.8: An undirected graphical model over random variables a, b, c, d and e. This graph corresponds to probability distributions that can be factored as

$$p(\mathrm{a}, \mathrm{b}, \mathrm{c}, \mathrm{d}, \mathrm{e}) = \frac{1}{Z}\phi^{(1)}(\mathrm{a}, \mathrm{b}, \mathrm{c})\phi^{(2)}(\mathrm{b}, \mathrm{d})\phi^{(3)}(\mathrm{c}, \mathrm{e}). \tag{3.56}$$

Figure 7: Example of an undirected graphical model.